BASIC CHEMISTRY

Module map

- Module 02: Basic Chemistry
- Connected lab: lab-02_probability-statistics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 02

Basic Chemistry

1. Atoms and elements
2. Bonding and molecular shape
3. Water as a biological solvent
4. pH and chemical balance
5. Chemical reactions in cells

Lab evidence

lab-02_probability-statistics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

BASIC CHEMISTRY

Learning objectives

- Use atomic structure to explain basic bonding patterns.
- Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Explain why water properties matter for cells and organisms.
- Interpret pH as a chemical condition that affects biology.
- Track atoms through a simple chemical reaction.

OBJECTIVE 1
Use atomic structure to explain basic bonding patterns.

OBJECTIVE 2
Distinguish covalent, ionic, and hydrogen bonding in biological examples.

OBJECTIVE 3
Explain why water properties matter for cells and organisms.

OBJECTIVE 4
Interpret pH as a chemical condition that affects biology.

OBJECTIVE 5
Track atoms through a simple chemical reaction.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

BASIC CHEMISTRY

Topic sequence

- Atoms and elements: Atoms form elements, and their electrons shape bonding behavior.
- Bonding and molecular shape: Covalent, ionic, and hydrogen bonds create different molecular properties.
- Water as a biological solvent: Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.
- pH and chemical balance: pH tracks hydrogen ion concentration and affects biological molecules.
- Chemical reactions in cells: Cells depend on chemical reactions that rearrange matter without destroying atoms.

01

Atoms and elements

Atoms form elements, and their electrons shape bonding behavior.

02

Bonding and molecular shape

Covalent, ionic, and hydrogen bonds create different molecular properties.

03

Water as a biological solvent

Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.

04

pH and chemical balance

pH tracks hydrogen ion concentration and affects biological molecules.

05

Chemical reactions in cells

Cells depend on chemical reactions that rearrange matter without destroying atoms.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

BASIC CHEMISTRY

Module 02: Basic Chemistry Evidence Map

- Module focus: Basic Chemistry
- Central claim: Basic Chemistry explains atomic structure with water, pH, and biological molecule behavior.
- Key nodes to connect: Atoms and elements, Bonding and molecular shape, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-02_probability-statistics.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

BASIC CHEMISTRY

Module 02: Basic Chemistry Reasoning Sequence

- Module focus: Basic Chemistry
- Trace the model: Atoms and elements -> Bonding and molecular shape -> Water as a biological solvent -> pH and chemical balance
- Inputs become outputs: Atoms and elements, Bonding and molecular shape -> Use atomic structure to explain basic bonding patterns., Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Feedback check: lab-02_probability-statistics.md evidence checks whether students can explain water as a biological solvent.
- Lab connection: lab-02_probability-

to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

BASIC CHEMISTRY

Terms as evidence handles

- Atom: Smallest unit of an element that keeps that element identity.
- Element: Pure substance defined by proton number.
- Covalent bond: Bond formed when atoms share electrons.
- Ionic bond: Attraction between oppositely charged ions.
- Hydrogen bond: Weak attraction involving polar hydrogen and electronegative atoms.
- Polarity: Uneven charge distribution across a molecule.

ATOM
Smallest unit of an element that keeps that element identity.

ELEMENT
Pure substance defined by proton number.

COVALENT BOND
Bond formed when atoms share electrons.

IONIC BOND
Attraction between oppositely charged ions.

HYDROGEN BOND
Weak attraction involving polar hydrogen and electronegative atoms.

POLARITY
Uneven charge distribution across a molecule.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

BASIC CHEMISTRY

Lab connection

- Lab file: lab-02_probability-statistics.md
- Lab evidence should test or illustrate: Water as a biological solvent
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Atoms and elements

EVIDENCE
lab-02_probability-statistics.md

CLAIM
Use atomic structure to explain basic bonding patterns.

Teaching note: Make the lab connection explicit before students encounter the activity.

BASIC CHEMISTRY

Contrast check

- Do not stop at: Atoms form elements, and their electrons shape bonding behavior.
- Stronger explanation: Cells depend on chemical reactions that rearrange matter without destroying atoms.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Atoms form elements, and their electrons shape bonding behavior.

DEEPER
Cells depend on chemical reactions that rearrange matter without destroying atoms.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

BASIC CHEMISTRY

Module 02: Basic Chemistry Retrieval and Lab Check

- Module focus: Basic Chemistry
- Retrieval prompt: How do protons, neutrons, and electrons differ?
- Answer check: Use atomic structure to explain basic bonding patterns.
- Required terms: Atom, Element, Covalent bond
- Lab connection: lab-02_probability-statistics.md; lab-02_probability-statistics.md supplies evidence for probability, graph reading, and chemical evidence claims.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

BASIC CHEMISTRY

Practice quiz bridge

- 1. Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom. What kind of bond holds the result together?
- 2. Why can water dissolve so many substances important to cells?
- 3. A solution changes from pH 7 to pH 4. This means it became:
- 4. Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?

QUESTION 1

Key: B

Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom. What kind of bond holds the result together?

QUESTION 2

Key: A

Why can water dissolve so many substances important to cells?

QUESTION 3

Key: D

A solution changes from pH 7 to pH 4. This means it became:

QUESTION 4

Key: C

Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?

Teaching note: Use the quiz as formative feedback. Answer key: 1=B, 2=A, 3=D, 4=C

BASIC CHEMISTRY

Synthesis and exit ticket

- Exit claim: explain atoms and elements using evidence from lab-02_probability-statistics.md.
- Must include: Atom, Element, Covalent bond.
- Revision prompt: what evidence would change your explanation?

CLAIM
Atoms and elements

EVIDENCE
lab-02_probability-statistics.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.